

7.11 Exercises

Exercise 7.1: QP with equality constraints

Obtain the solution u^0 and the value V^0 of the equality constrained optimization problem $V^0 = \min_u \{V(u) \mid h(u) = 0\}$ where $V(u) = (1/2)u'Ru + r'u + c$ and $h(u) := Mu - p$.

Exercise 7.2: Parametric QP with equality constraints

Show that the solution $u^0(x)$ and the value $V^0(x)$ of the parametric optimization problem $V^0(x) = \min_u \{V(x, u) \mid h(x, u) = 0\}$ where $V(x, u) := (1/2)x'Qx + u'Sx + (1/2)u'Ru + q'x + r'u + c$ and $h(x, u) := Mu - Nx - p$ have the form $u^0(x) = Kx + k$ and $V^0(x) = (1/2)x'\tilde{Q}x + \tilde{q}'x + s$. Determine $\tilde{Q}$, $\tilde{q}$, s , K , and k .

Exercise 7.3: State and input trajectories in constrained LQ problem

For the constrained linear quadratic problem defined in Section 7.4, show that $\mathbf{u} := (u(0), u(1), \dots, u(N-1))$ and $\mathbf{x} := (x(0), x(1), \dots, x(N))$, where $x(0) = x$ and $x(i) = \phi(i; x, \mathbf{u})$, $i = 0, 1, \dots, N$, satisfy

$$\mathbf{x} = \mathbf{F}x + \mathbf{G}\mathbf{u}$$

and determine the matrices $\mathbf{F}$ and $\mathbf{G}$; in this equation $\mathbf{u}$ and $\mathbf{x}$ are column vectors. Hence show that $V_N(x, \mathbf{u})$ and $\mathbb{Z}$, defined respectively in (7.9) and (7.11), satisfy (7.12) and (7.13), and determine $\mathbf{Q}$, $\mathbf{R}$, $\mathbf{M}$, $\mathbf{N}$, and $\mathbf{p}$.

Exercise 7.4: The parametric LP with unique minimizer

For the example of Figure 7.7, determine $u^0(x)$, $V^0(x)$, $I^0(x)$, and $C^*(x)$ for all x in the interval $[x_1, x_3]$. Show that $-r$ lies in $C^*(x)$ for all x in $[x_1, x_3]$.

Exercise 7.5: Cost function and constraints in constrained LQ control problem

For the constrained linear control problem considered in Section 7.8, determine the matrices $\mathbf{M}$, $\mathbf{N}$, and $\mathbf{p}$ that define the constraint set $\mathbb{Z}$, and the vectors $\mathbf{q}$ and $\mathbf{r}$ that define the cost $V_N(\cdot)$.

Exercise 7.6: Cost function in constrained linear control problem

Show that $|x|_p$, $p = 1$ and $p = \infty$, may be expressed as $\max_j \{s'_j x \mid j \in J\}$ and determine s_i , $i \in I$ for the two cases $p = 1$ and $p = \infty$. Hence show that the optimal control problem in Section 7.8 may be expressed as

$$V_N^0(x) = \min_{\mathbf{v}} \{V_N(x, \mathbf{v}) \mid \mathbf{M}\mathbf{v} \leq \mathbf{N}x + \mathbf{p}\}$$

where, now, $\mathbf{v}$ is a column vector whose components are $u(0), u(1), \dots, u(N-1), \ell_x(0), \ell_x(1), \dots, \ell_x(N), \ell_u(0), \ell_u(1), \dots, \ell_u(N-1)$ and f ; the cost $V_N(x, \mathbf{v})$ is now defined by

$$V_N(x, \mathbf{v}) = \sum_{i=0}^{N-1} (\ell_x(i) + \ell_u(i)) + f$$

Finally, $\mathbf{M}\mathbf{v} \leq \mathbf{N}\mathbf{x} + \mathbf{p}$ now specifies the constraints $\mathbf{u}(i) \in \mathbb{U}$ and $\mathbf{x}(i) \in \mathbb{X}$, $|\mathbf{R}\mathbf{u}(i)|_p \leq \ell_u(i)$, $|\mathbf{Q}\mathbf{x}(i)|_p \leq \ell_x(i)$, $i = 0, 1, \dots, N-1$, $\mathbf{x}(N) \in \mathbb{X}_f$, and $|\mathbf{Q}_f\mathbf{x}(N)| \leq f$. As before, $\mathbf{x}^+ = \mathbf{F}\mathbf{x} + \mathbf{G}\mathbf{u}$.

Exercise 7.7: Is QP constraint qualification relevant to MPC?

Continuity properties of the MPC control law are often used to establish robustness properties of MPC such as robust asymptotic stability. In early work on continuity properties of linear model MPC, Scokaert, Rawlings, and Meadows (1997) used results on continuity of QPs with respect to parameters to establish MPC stability under perturbations. For example, Hager (1979) considered the following QP

$$\min_{\mathbf{u}} (1/2) \mathbf{u}' \mathbf{H} \mathbf{u} + \mathbf{h}' \mathbf{u} + c$$

subject to

$$\mathbf{D}\mathbf{u} \leq \mathbf{d}$$

and established that the QP solution $\mathbf{u}^0$ and cost V^0 are Lipschitz continuous in the data of the QP, namely the parameters $\mathbf{H}, \mathbf{h}, \mathbf{D}, \mathbf{d}$. To establish this result Hager (1979) made the following assumptions.

- The solution is unique for all $\mathbf{H}, \mathbf{h}, \mathbf{D}, \mathbf{d}$ in a chosen set of interest.
- The rows of $\mathbf{D}$ corresponding to the constraints active at the solution are linearly independent. The assumption of linear independence of active constraints is a form of *constraint qualification*.

- (a) First we show that some form of constraint qualification is required to establish continuity of the QP solution with respect to matrix $\mathbf{D}$. Consider the following QP example that does not satisfy Hager's constraint qualification assumption.

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \quad \mathbf{d} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \mathbf{h} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \quad c = 1$$

Find the solution $\mathbf{u}^0$ for this problem.

Next perturb the $\mathbf{D}$ matrix to

$$\mathbf{D} = \begin{bmatrix} 1 & 1 \\ -1 + \epsilon & -1 \end{bmatrix}$$

in which $\epsilon > 0$ is a small perturbation. Find the solution to the perturbed problem. Are V^0 and $\mathbf{u}^0$ continuous in parameter $\mathbf{D}$ for this QP? Draw a sketch of the feasible region and cost contours for the original and perturbed problems. What happens to the feasible set when $\mathbf{D}$ is perturbed?

- (b) Next consider MPC control of the following system with state inequality constraint and no input constraints

$$\mathbf{A} = \begin{bmatrix} -1/4 & 1 \\ -1 & 1/2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \quad \mathbf{x}(k) \leq \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad k \in \mathbb{I}_{0:N}$$

Using a horizon $N = 1$, eliminate the state $\mathbf{x}(1)$ and write out the MPC QP for the input $\mathbf{u}(0)$ in the form given above for $\mathbf{Q} = \mathbf{R} = \mathbf{I}$ and zero terminal penalty. Find an initial condition $\mathbf{x}_0$ such that the MPC constraint matrix $\mathbf{D}$ and vector $\mathbf{d}$ are identical to those given in the previous part. Is this $\mathbf{x}_0 \in \mathcal{X}_N$?

Are the rows of the matrix of active constraints linearly independent in this MPC QP on the set $\mathcal{X}_N$? Are the MPC control law $\kappa_N(\mathbf{x})$ and optimal value function $V_N^0(\mathbf{x})$ Lipschitz continuous on the set $\mathcal{X}_N$ for this system? Explain the reason if these two answers differ.

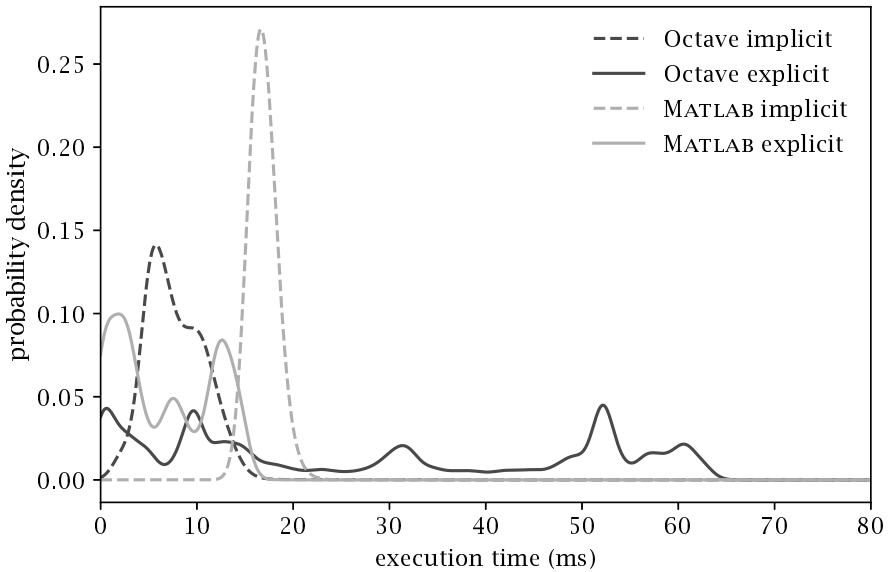


Figure 7.8: Solution times for explicit and implicit MPC for $N = 20$. Plot shows kernel density estimate for 10,000 samples using a Gaussian kernel ($\sigma = 1$ ms).

Exercise 7.8: Explicit versus implicit

Using the system from Figure 7.6, find the explicit control law for horizon $N = 20$ (you should find 1719 regions). Implement a simple lookup function for the explicit control law. Randomly sample a large number of points (≥ 1000) from X_N and compare execution times for explicit MPC (via the lookup function) and implicit MPC (via solving a QP). Which method is better? Example results are shown in Figure 7.8, although your times may vary significantly. How could you improve your lookup function?

Exercise 7.9: Cascaded MPC and PID

Consider a Smart TankTM of liquid whose height h evolves according to

$$\tau \frac{dh}{dt} + h = Kq, \quad \tau = 10, K = 1$$

with q the (net) inflow. The tank is SmartTM in that it has an integrated PI controller that computes

$$q = K_c \left(h_{sp} - h + \frac{1}{\tau_c} \epsilon \right)$$

$$\epsilon = \int h_{sp} - h \, dt$$

so that the height of the tank returns to h_{sp} automatically. Unfortunately, the controller parameters are not very SmartTM, as they are fixed permanently at $K_c = 1/2$ and $\tau_c = 1$.

- (a) Simulate the closed-loop behavior of the system starting from $h = -1$, $\epsilon = 0$ with $h_{\text{sp}} = 0$.
- (b) Design an MPC controller to choose h_{sp} . As a cost function take

$$\ell(h, \epsilon, q, h_{\text{sp}}) = 5(h^2 + \epsilon^2) + q^2 + 10h_{\text{sp}}^2$$

so that the controller drives the system to $h = \epsilon = 0$. Choose $\Delta = 1$. How does performance compare to the previous case? How much storage (i.e., how many floating-point numbers must be stored) to implement this controller?

- (c) Add the constraint $q \in [-0.2, 0.2]$ to the MPC formulation, and design an explicit MPC controller valid for $h \in [-5, 5]$ and $\epsilon \in [-10, 10]$ (use `solvmppq.m` from Figure 7.6, and add constraints $Ep \leq e$ to only search the region of interest). How large does N have to be so that the full region is covered? How much storage is needed to implement this controller?

Exercise 7.10: Explicit economic MPC for electricity arbitrage

Electricity markets are often subject to real-time pricing, whereby the cost of purchasing electricity varies with time. Suppose that you have a large battery that allows you to buy electricity at one time and then sell it back to the grid at another. We can model this as a simple integrator system

$$x^+ = x + u$$

with x representing the amount of stored energy in the tank, and u giving the amount of electricity that is purchased for the battery $u > 0$ or discharged from the battery and sold back to the grid ($u < 0$). We wish to find an explicit control law based on the initial condition $x(0)$ a known forecast of electricity prices $c(0), c(1), \dots, c(N-1)$.

- (a) To start, suppose that u is constrained to the interval $[-1, 1]$ but x is unconstrained. A reasonable optimization problem is

$$\begin{aligned} \min_u \quad & \sum_{k=0}^{N-1} c(k)u(k) + 0.1u(k)^2 \\ \text{s.t.} \quad & x(k+1) = x(k) + u(k) \\ & u(k) \in [-1, 1] \end{aligned}$$

where the main component of the objective function is the cost of electricity purchase/sale with a small penalty added to discourage larger transactions. By removing the state evolution equation, formulate an explicit quadratic programming problem with N variables (the $u(k)$) and $N+1$ parameters ($x(0)$ and the price forecast $c(k)$). What is a theoretical upper bound on the number of regions in the explicit control law? Assuming that $x(0) \in [-10, 10]$ and each $c(k) \in [-1, 1]$, find the explicit control law for a few small values of N . (Consider using `solvmppq.m` from Figure 7.6; you will need to add constraints $Ep \leq e$ on the parameter vector to make sure the regions are bounded.) How many regions do you find?

- (b) To make the problem more realistic, we add the constraint $x(k) \in [-10, 10]$ to the optimization, as well as an additional penalty on stored inventory. The optimization problem is then

$$\begin{aligned} \min_{\mathbf{u}} \quad & \sum_{k=0}^{N-1} c(k)u(k) + 0.1u(k)^2 + 0.01x(k)^2 \\ \text{s.t.} \quad & x(k+1) = x(k) + u(k) \\ & u(k) \in [-1, 1] \\ & x(k) \in [-10, 10] \end{aligned}$$

Repeat the previous part but using the new optimization problem.

- (c) Suppose you wish to solve this problem with a 7-day horizon and a 1-hour time step. Can you use the explicit solution of either formulation? (Hint: for comparison, there are roughly 10^{80} atoms in the observable universe.)

Bibliography

- A. Alessio and A. Bemporad. A survey on explicit model predictive control. In L. Magni, D. Raimondo, and F. Allgöwer, editors, *Nonlinear Model Predictive Control - Towards New Challenging Applications*, pages 345–369. Springer Berlin / Heidelberg, 2009.
- B. Bank, J. Guddat, D. Klatte, B. Kummer, and K. Tanner. *Non-linear parametric optimization*. Birkhäuser Verlag, Basel, Boston, Stuttgart, 1983.
- A. Bemporad, M. Morari, V. Dua, and E. N. Pistikopoulos. The explicit linear quadratic regulator for constrained systems. *Automatica*, 38(1):3–20, 2002.
- F. Borrelli. *Constrained Optimal Control of Linear and Hybrid Systems*. Springer-Verlag Berlin Heidelberg, 2003.
- F. Borrelli, A. Bemporad, and M. Morari. *Predictive Control for Linear and Hybrid Systems*. Cambridge University Press, 2017.
- S. Columbano, K. Fukuda, and C. N. Jones. An output sensitive algorithm for multiparametric LCPs with sufficient matrices. *Polyhedral Computation*, 48: 73, 2009.
- G. B. Dantzig, J. Folkman, and N. Z. Shapiro. On the continuity of the minimum set of a continuous function. *Journal of Mathematical Analysis and Applications*, 17(3):519–548, 1967.
- C. Feller, T. A. Johansen, and S. Olaru. An improved algorithm for combinatorial multi-parametric quadratic programming. *Automatica*, 49(5):1370–1376, 2013.
- T. Gal and J. Nedoma. Multiparametric linear programming. *Management Science*, 18(7):406–422, 1972.
- S. I. Gass and T. L. Saaty. The computational algorithm for the parametric objective function. *Naval Research Logistics Quarterly*, 2:39–45, 1955.
- W. W. Hager. Lipschitz continuity for constrained processes. *SIAM J. Cont. Opt.*, 17(3):321–338, 1979.
- M. Herceg, M. Kvasnica, C. N. Jones, and M. Morari. Multi-Parametric Toolbox 3.0. In *Proc. of the European Control Conference*, pages 502–510, Zürich, Switzerland, July 17–19 2013. <http://control.ee.ethz.ch/~mpt>.

- C. N. Jones. Approximate receding horizon control. In F. Borrelli, A. Bemporad, and M. Morari, editors, *Predictive Control for Linear and Hybrid Systems*, pages 277–300. Cambridge University Press, 2017.
- C. N. Jones, M. Barić, and M. Morari. Multiparametric linear programming with applications in control. *Eur. J. Control*, 13:152–170, 2007.
- D. Q. Mayne and S. V. Raković. Optimal control of constrained piecewise affine discrete-time systems using reverse transformation. In *Proceedings of the IEEE 2002 Conference on Decision and Control*, volume 2, pages 1546 – 1551 vol.2, Las Vegas, USA, 2002.
- D. Q. Mayne and S. V. Raković. Optimal control of constrained piecewise affine discrete-time systems. *Comp. Optim. Appl.*, 25(1-3):167–191, 2003.
- D. Q. Mayne, S. V. Raković, and E. C. Kerrigan. Optimal control and piecewise parametric programming. In *Proceedings of the European Control Conference 2007*, pages 2762–2767, Kos, Greece, July 2–5 2007.
- P. O. M. Scokaert, J. B. Rawlings, and E. S. Meadows. Discrete-time stability with perturbations: Application to model predictive control. *Automatica*, 33(3): 463–470, 1997.
- M. M. Serón, J. A. De Doná, and G. C. Goodwin. Global analytical model predictive control with input constraints. In *Proceedings of the 39th IEEE Conference on Decision and Control*, pages 154–159, Sydney, Australia, December 2000.